

Pin Desk

An autonomous options desk that maps dealer gamma to find where the market gets pinned — and knows when not to trade.

DEALER GAMMA

12 RISK GATES

LLM CAN ONLY SHRINK

ALPACA TRADING API + CLI

It doesn't predict direction. It finds where dealers pin the market.

THE PROBLEM

Over four trading sessions, direction is a coin flip.

Most entrants in this event are building the same thing: an LLM that picks a stock, forms an opinion, and buys a call. That's a lottery ticket dressed up as a strategy — and judges score P&L, not conviction.

~4

trading sessions before judging

15 min

free-feed options delay

0

reliance on that delayed feed

WHAT WE BUILT INSTEAD

- A strategy that's direction-agnostic — it trades a regime, not a guess
- A primary input (open interest) that updates once a day, immune to the 15-min feed delay
- A risk kernel with veto power — not an LLM asked nicely to be careful
- Standing down is a real answer, not a failure state

THE THESIS

Dealers hedge. Hedging moves price. Price gets pinned.

Market makers are short the options retail is long. To stay hedged, they buy weakness and sell strength — damping price near the strikes carrying the heaviest open interest.

POSITIVE GAMMA

Dealers are long gamma. Hedging flow leans against the move — buy dips, sell rips. The tape holds a range.

Sell defined-risk premium around the pin.

NEGATIVE GAMMA

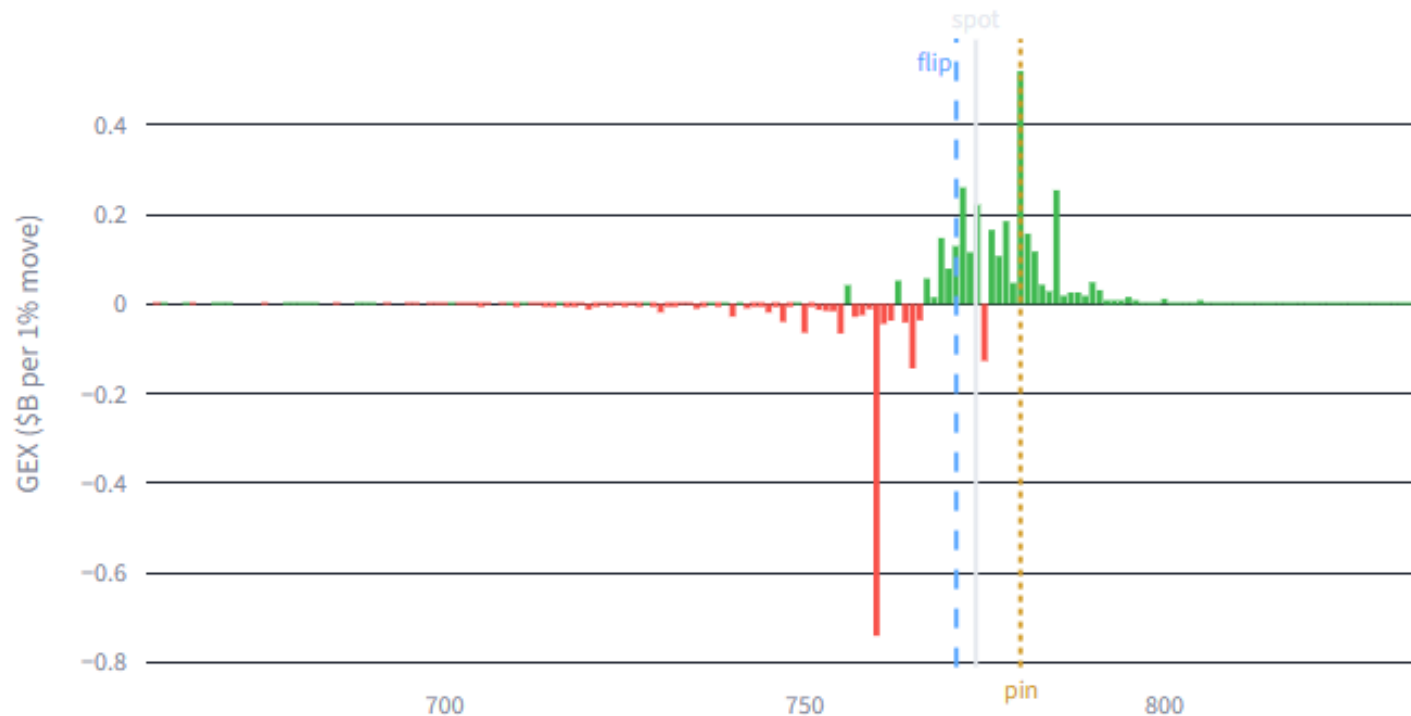
Dealers are short gamma. Hedging flow leans with the move — sell into weakness, buy into strength. The tape trends.

Stand down. No trade, not a smaller one.

Real chain. Real flip point. Real pin.

Net dealer gamma per strike, computed as $\text{open interest} \times \text{gamma} \times 100 \times \text{spot}^2 \times 1\%$, then re-priced with Black-Scholes across a swept spot range to interpolate the exact zero crossing.

SPY — dealer gamma exposure by strike



- Damps moves
- Amplifies moves
- Flip / pin / spot

2026-08-27 • SPY

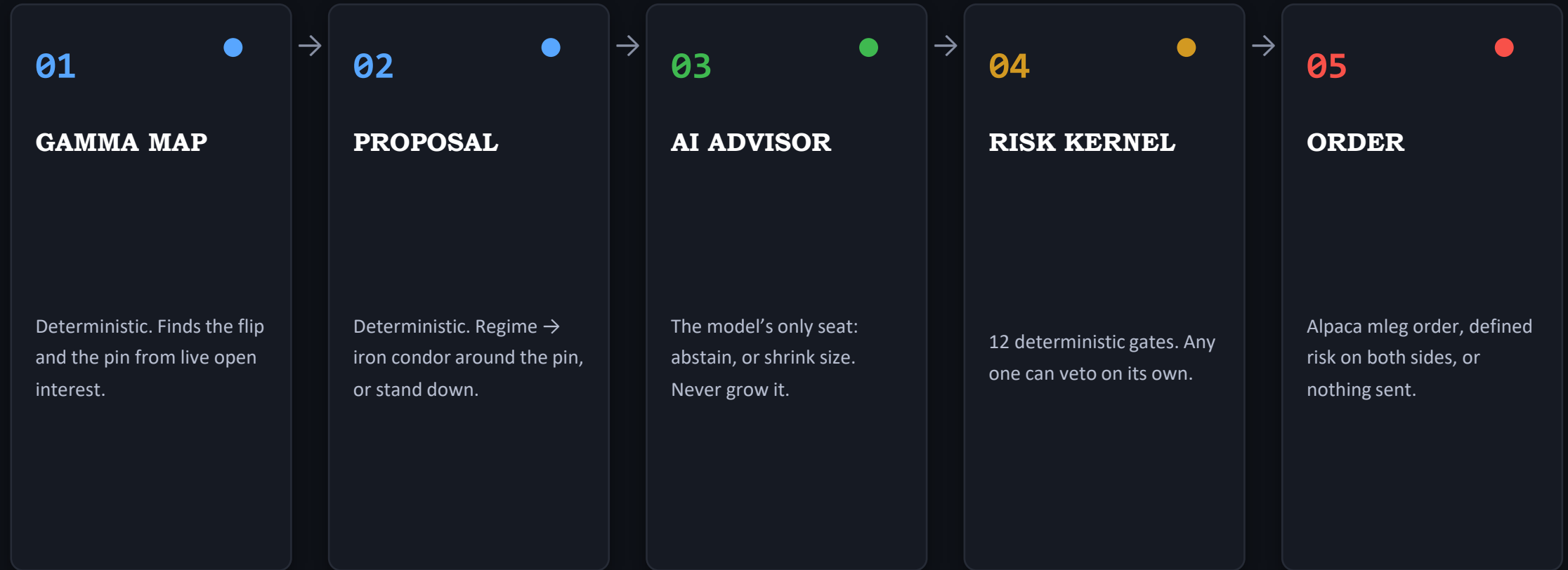
Spot 772.23

Flip 768.83

Pin 780.00

Regime **POSITIVE**

Five stages. The model gets exactly one of them.



Exits run before entries every cycle — closing frees risk budget the gates would otherwise refuse.

The model can only make a trade smaller. Never bigger.

Its entire action space: abstain, or a conviction dial in $[0, 1]$ that scales position size. It cannot choose strikes, move an expiration, or change the structure — and the clamp is enforced twice, schema and code.

“An LLM asked to be careful is not a risk control. A gate that rejects the order is.”

If the model is unreachable, slow, or malformed, the desk trades deterministically and logs that it did — verified live against a genuine API failure.

LIVE EXAMPLE · QQQ

Deterministic size **2 contracts**

Advisor stance **reduce**

Conviction **0.60**

Final size **1 contract**

ADVISOR RATIONALE (LOGGED, VERBATIM)

“The short strikes are ~2% away from spot... open interest is healthy on all legs... credit is modest and the short call is near the pin.”

Twelve gates. Any one can veto. All twelve always run.

Gates don't short-circuit, so the journal shows how badly a trade failed — not just that it did.

- **sanity**

structure or credit is invalid

- **regime**

not in positive gamma

- **flip_margin**

spot within 0.25% of the flip

- **defined_risk**

any uncovered short leg

- **duplicate**

same underlying + expiry held

- **tenor**

outside 5–12 DTE

- **data_freshness**

expiration already passed

- **credit_quality**

credit < 15% of width

- **liquidity**

OI < 250 or spread > 25%

- **position_size**

> 1% NAV in one trade

- **portfolio_risk**

> 6% NAV across the book

- **concentration**

> 2 positions per name

- *found live: the desk doubled its own position, then read a knife-edge regime as decided*

Trading API, Market Data API, and the official CLI — not REST alone.



TRADING API

Account state, option-contract metadata — open interest lives here, not on the market-data snapshot — and mleg order construction. Verified live: a condor filled better than its limit, both open and close.



MARKET DATA API

Option snapshots — greeks, IV, quotes — joined against Trading-API contracts by symbol to build the gamma map. The 15-min-delayed free feed never binds this strategy.



OFFICIAL CLI

alpaca position list is the authoritative source for reconciliation every cycle — a separate process, separate auth, separate connection. A REST client mid-outage can't quietly agree with itself.

Auth by environment variable, deliberately — alpaca profile login opens a browser, which a scheduled loop can't do.

PROOF IT WORKS

It traded autonomously. We verified every step.

A full four-leg QQQ iron condor, opened and closed on paper with zero human intervention — cross-checked through two independent paths that had to agree.

OPEN

limit -1.19

filled -1.23

Better than the limit — confirms
Alpaca's negative-for-credit
convention

VERIFY

REST client

Alpaca CLI

Two independent processes, separate
auth — both agreed: 4 legs held

CLOSE

limit +1.46

filled +1.42

Also better than the limit; book
returned to flat

RESULT

entry \$1.32

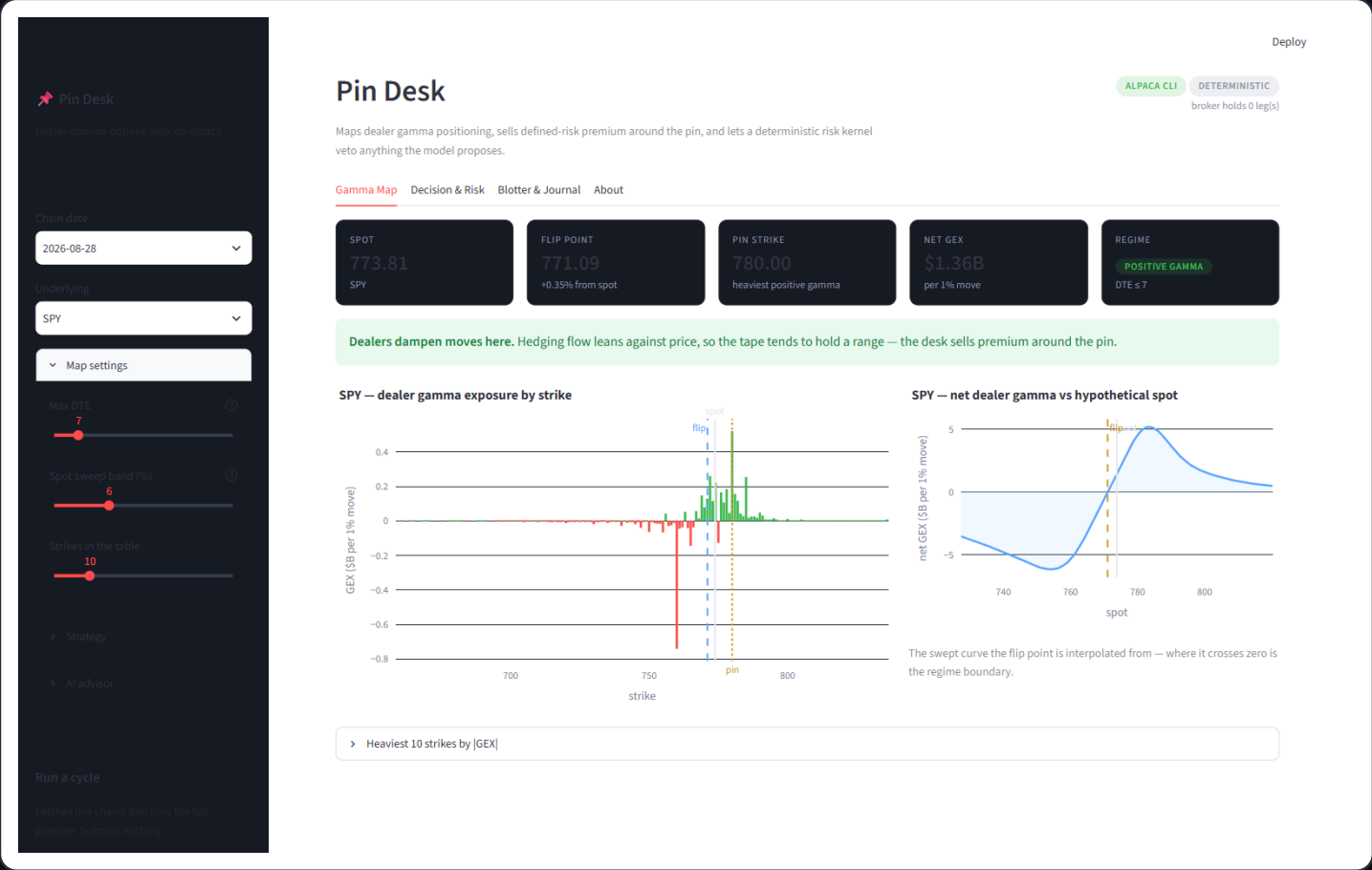
exit \$1.42

-\$10 round trip — the bid/ask drag
the exit model predicts

Account PA36VRS31NZ5 (dev) · order ids logged · fills confirmed via alpaca.exe position list and REST get_all_positions()

Same numbers, same code path — just a live view.

The dashboard has no trading logic of its own. Every figure on screen calls straight into the same gamma, strategy, risk, and advisor modules the agent loop runs — it cannot drift from what the desk actually does.



WHERE THINGS STAND

Ready for the live window.

● Competition account	PA3HAQRS59A5	<i>verified \$100,000.00, zero prior orders</i>
● Trading window	Mon 31 Aug – Thu 3 Sep	<i>book flattens before Fri judging opens</i>
● Test coverage	183 tests	<i>clamp, gates, exits, orders, CLI fallback</i>
● Data history	recording daily	<i>open interest can't be backfilled once missed</i>

Pin Desk

Team Gasbin · Alpaca AI Trading Agents Hackathon 2026